FIRST TERM CONTINUOUS ASSESSMENT 1 2026/2027

SUBJECT: MATHEMATICS	**SECTION(S):**A	**11TH GRADE**

TIME ALLOWED: 20MINUTES

SECTION A: OBJECTIVES

1. If $\frac{6x-y}{x+2y} = 2$, find the value of $\frac{x}{y}$.
   A. $\frac{5}{8}$
   B. $\frac{4}{5}$
   C. $\frac{5}{4}$
   D. $\frac{8}{5}$

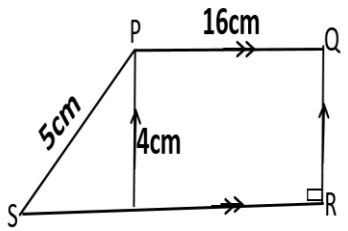
2. Given that $81 \times 2^{2n-2}$ = k, find the $\sqrt{k}$.
   A. $4.5 \times 2^{2n-2}$
   B. $9 \times 2^{n-1}$
   C. 9×2^{2n}
   D. 4.5×2^{2n}

3. The implication $X \rightarrow Y$ is equivalent to?
   A. $\sim X \rightarrow \sim Y$
   B. $\sim Y \rightarrow \sim X$
   C. $Y \rightarrow \sim X$
   D. $Y \rightarrow X$

4. If $P \rightarrow Q$ is False only when.....
   A. P is true and Q is true
   B. P is true and Q is false
   C. P is false and Q is true
   D. P is false and Q is false

5. Given that one of the roots of the equation $2x^2 + (k + 2)x + k = 0$ is 2, find the value of k.
   A. -4
   B. -2
   C. -1
   D. $-\frac{1}{4}$

6. Calculate the perimeter of the trapezium, PQRS.
   A. 24cm
   B. 36cm
   C. 44cm
   D. 49cm



7. Find the number whose logarithms to base 10 is 2.6025
   A. 400.4
   B. 0.4004
   C. 0.04004
   D. 0.004004

8. If $\log 2 = 0.3010$ and $\log 2^y = 1.8062$. find, correct to the nearest whole number, the value of y
   A. 6
   B. 5
   C. 4
   D. -5

9. The foot of a ladder is 6m from the base of electric pole. The top of the ladder rests against the pole at a point 8m above ground. Calculate to the nearest degree the angle the ladder makes with the pole.
   A. 25^0
   B. 34^0
   C. 37^0
   D. 43^0

10. Without using tables, find the value of $\frac{sin20^o}{cos70^o} + \frac{cos25^o}{sin65^o}$
    A. 2
    B. 1
    C. 0
    D. -1

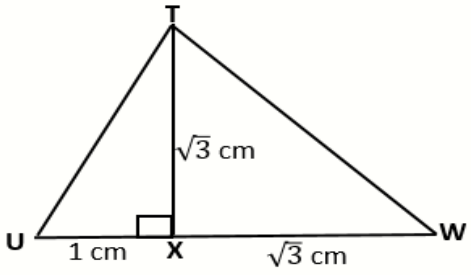
11. Halima is n years old. Her brother's age is 5 years more than half of her age. How old is her brother?
    A. 2n + 5
    B. $\frac{1}{2} + 5n$
    C. $\frac{n}{2} - 5$
    D. $\frac{n}{2} + 5$

12. If $\sin(x + 30)^o = cos40^o$, find the value of x.

A. 10°
B. 20°
C. 30°
D. 40°

13. A ladder leans against a vertical wall at an angle of 60° to the wall. If the foot of the ladder is 5 meters away from the wall, calculate the length of the ladder.
A. $\frac{10\sqrt{3}}{3}m$
B. $5m$
C. $5\sqrt{3}m$
D. 10m

14. In the diagram, TX is perpendicular to UW, |UX| = 1 cm and TX = WX = $\sqrt{3}$ cm. Calculate the size of $\angle UTW$.



A. 135^0
B. 105^0
C. 75^0
D. 60^0

15. Which of following is a valid conclusion from the premise. "Nigeria footballers are good footballers"?
A. Joseph plays football in Nigeria therefore he is a good footballer
B. Joseph is a good footballer therefore he is a Nigerian footballer
C. Joseph is a Nigerian footballer therefore he is a good footballer
D. Joseph plays good football therefore he is a Nigerian footballer

16. If $\sin\theta = k$, find $\tan\theta$, $0^0 \leq \theta \leq 90^o$
A. $1 - k$
B. $\frac{k}{\sqrt{1-k^2}}$
C. $\frac{k}{\sqrt{1-k}}$
D. $\frac{k}{\sqrt{k^2-1}}$

17. Find the area of an equilateral triangle of side 16cm.
A. $64\sqrt{3}$ cm^2
B. $72\sqrt{3}$ cm^2
C. 96 cm^2
D. $128\sqrt{3}$ cm^2

18. A sector of a circle of radius 7.2cm which subtends an angle of 300° at the centre is used to form a cone. What is the radius of the base of the cone?
A. 8cm
B. 6cm
C. 9cm
D. 7cm

19. Without using tables, evaluate $(343)^{\frac{1}{3}} \times (0.14)^{-1} \times (25)^{-\frac{1}{2}}$
A. 10
B. 12
C. 8
D. 7

20. A,B and C are subsets of universal set U such that U={0,1,2,3……..11,12}, A={x:0<x<7}, B={4,6,8,10}, C={1<x<8}. Find $(A \cup C)'$
A. {0,1,9}
B. {2,3,4,5,6}
C. {2,3,5,7,8}
D. {0,1,2,9,10}